Kite & Trapezium — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1 **(a) Trapezium.** It has one pair of parallel sides.

Q2 **(c) intersect at right angles.** A kite's diagonals are perpendicular.

Q3 **75.** $\angle P + \angle S = 180° \Rightarrow 180 - 105 = 75$.

Q4 **(b) rhombus.** A kite that is also a parallelogram has all four sides equal.

Q5 **True.** The equal non-parallel sides are what make it isosceles.

Q6 **(d) A false, R true.** Not every kite is a rhombus, though a rhombus does have all sides equal.

Short answer

Q7 **$\angle R = 80°$, $\angle S = 60°$.** Co-interior angles on each leg are supplementary.

Q8 **$\angle A = \angle C = 110°$.** $360 - 100 - 40 = 220$, shared equally between the equal pair.

Q9 **Any two of:** two pairs of adjacent equal sides; diagonals perpendicular; main diagonal bisects the other and bisects two corner angles; one pair of opposite angles equal.

Long answer

Q10 **$\angle C = 70°$, $\angle D = 105°$.** $AB \parallel DC \Rightarrow \angle A + \angle D = 180°$ and $\angle B + \angle C = 180°$.

Q11 **70°, 110°, 110°.** Base angles equal (the other base angle is 70°); each top angle = $180 - 70 = 110°$.

Q12 **(i) Square. (ii) Yes** — a parallelogram has a pair of parallel sides, so it fits the trapezium definition. **(iii) No** — a kite has only two pairs of adjacent equal sides, not all four.

Total: 21 marks.